

ME 465 CHEATSHEET | MATHEMATICAL CONTEXT

Summation Convention: $(u_i v_i)^2 = (\sum_i u_i v_i)^2$
 Kronecker Delta: $\delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$ $\delta_{ij} = e_j \cdot e_i$

$u \cdot v = (u_i e_i) \cdot (v_j e_j) = u_i v_j (e_i \cdot e_j) = u_i v_j \delta_{ij} = |u_i v_i|$
 (this is a scalar)

Tricky Relationship: $(v_j \delta_{ij}) e_i = v_i e_i = v$

TENSOR PRODUCTS $(a \otimes b) = a_i b_j e_i \otimes e_j$

$(a \otimes b) \cdot c = (b \cdot c) a$ $(a \otimes b)(c \otimes d) = (b \cdot c) a \otimes d$

Using tensor products as a basis for constructing matrices

$A = A_{ij} (e_i \otimes e_j) \iff A_{ij} = e_i \cdot A e_j$
 ($\Sigma \Sigma$ of basis matrices)

Euclidean Point Space (E^3) vs Vector space (E_3)

Gradient, divergence, curl (∇) maps to higher order

$\nabla \phi = \phi_{,i} e_i = \frac{\partial \phi}{\partial x_i} e_i$ ($\nabla \cdot$) maps to lower order
 $\nabla v = v_{,ij} e_i \otimes e_j$ ($\nabla \times$) maps ONLY $v \rightarrow \omega$

$\nabla A = A_{,ijk} e_i \otimes e_j \otimes e_k$

$\text{div } v = v_{,i} = \sum \frac{\partial v_i}{\partial x_i}$

$\text{div } A = A_{,ijj} e_i = \sum \frac{\partial}{\partial x_j} A_{ij} e_i$

$\text{curl } v = e_{ijk} v_{,kj} e_i$ $d v = (\nabla v) dx$

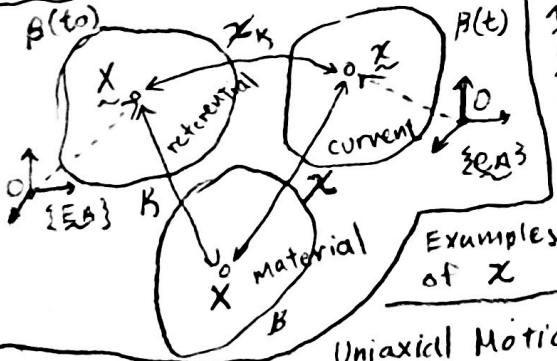
$\text{div}(\nabla \phi) = \nabla^2 \phi$ LAPLACIAN

DIRECTIONAL DERIV. of $\phi(x)$ along u @ point x_0
 $= |\nabla \phi(x)|_{x_0} \cdot u$

MATERIAL, REFERENCE, & SPATIAL

$\underline{x} = \underline{\chi}(\underline{X}) = \underline{\chi} \underline{\kappa}^{-1}(\underline{x}, t)$

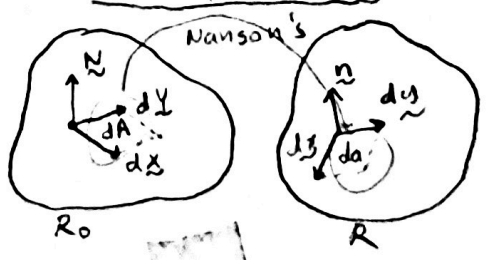
$\underline{\kappa} = \underline{\chi}(\underline{X}, t) = \underline{\chi} \underline{\kappa}(\underline{X}, t)$



Simple Shearing: $x_1 = X_1 + \alpha(t) X_2$
 $x_2 = X_2$
 $x_3 = X_3$

Nanson's Formula for Area
 $n da = d\tilde{x} \times d\tilde{y} = F^* (d\tilde{x} \times d\tilde{y})$
 $= F^* N dA$

Intuition for Area & Volume Transformations



Box Product: $[u, v, w] = u \cdot v \times w$

// used to find volume of parallel-piped

Permutation Symbol: $e_{ijk} \equiv e_i \cdot e_j \times e_k$

$e_{ijk} = \begin{cases} 0 & \text{if any 2 } i, j, k \text{ are equal} \\ 1 & \text{if } ijk = 1, 2, 3 \text{ or } 3, 2, 1 \\ -1 & \text{if } ijk = 1, 3, 2 \text{ or } 2, 3, 1 \end{cases}$

Intuitive $e_i \times e_j = e_{ijk} e_k$ $u \times v = e_{ijk} u_i v_j e_k$

$[u, v, w] = e_{ijk} u_i v_j w_k$

TENSOR PROPERTIES

Product of Tensors

- $(AB) \cdot c = A(B \cdot c)$
- $A \cdot B = A_{ik} B_{kj} e_i \otimes e_j$
- $AB \neq BA$ in general

Inverse of Tensors

- $AA^{-1} = A^{-1}A = I$
- $A^{-1} = A_{ji}^{-1} e_j \otimes e_i$
- $(A^{-1})^{-1} = A$
- A is invertible iff $\det A \neq 0$ $A^{-1} = -A$ or $(A_{ij} = -A_{ji})$

Orthogonal Tensors

- $Q Q^T = Q^T Q = I$
- $Q^{-1} = Q^T$
- $\det Q = \pm 1$
- $Q u \cdot Q v = u \cdot v$

OTHER PROPERTIES

- $(AB)^{-1} = B^{-1} A^{-1}$
- $(A^T)^{-1} = (A^{-1})^T$
- $\det(A^T) = \det(A)$
- $\det(A^{-1}) = 1/\det(A)$
- $\det(AB) = \det(A) \det(B)$
- $\frac{d}{dt}(AB) = \frac{dA}{dt} B + \frac{dB}{dt} A$

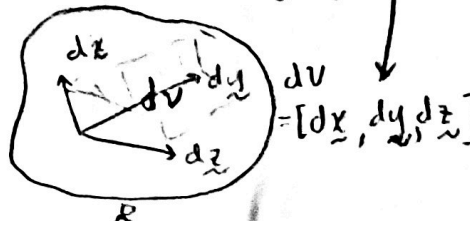
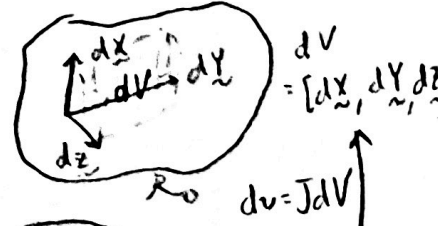
MATERIAL DERIVATIVE

Jacobian Proof that $dv = J dV$

$dv = [dx, dy, dz] = [F dx, F dy, F dz] = \det F [dx, dy, dz] = J dV$

"Isochoric": volume does not change between curr $\rightarrow$ ref.

$J = 1 \implies dv = dV \implies \rho_k = \rho_J$



Proof that $\underline{C} = \underline{U}^T \underline{A} \underline{U}$ & $\underline{B} = \underline{V}^T \underline{A} \underline{V}$

$$\underline{C} = \underline{F}^T \underline{F} = (\underline{R} \underline{U})^T (\underline{R} \underline{U}) = \underline{U}^T \underline{R}^T \underline{R} \underline{U} = \underline{U}^T \underline{U} = \underline{U}^T \underline{U} = \underline{U}^T \underline{U}$$

since $\underline{U}$ symmetric

Positive Definiteness $\triangleq$ for all $\underline{a} \neq \underline{0}$, $\underline{a} \cdot \underline{A} \underline{a} > 0$
 Implies invertible, eigvals > 0 , $\det \underline{A} > 0$

L & D IDENTITIES

- ① $\dot{\underline{C}} = 2 \underline{F}^T \underline{D} \underline{F}$ ② $\dot{\underline{E}} = \underline{F}^T \underline{D} \underline{F}$
- ③ $\dot{\underline{B}} = \underline{L} \underline{B} + \underline{B} \underline{L}^T$ ④ $\dot{\underline{E}} = \frac{1}{2} (\underline{B}^{-1} \underline{L} + \underline{L}^T \underline{B}^{-T})$
- ⑤ $\frac{d}{dt} \ln \mu = \underline{m} \cdot \underline{D} \underline{m}$

Derivation of RTT Motivator: $\frac{d}{dt} \int_P \tilde{\phi} dv$

$$\frac{d}{dt} \int_P \tilde{\phi} dv = \frac{d}{dt} \int_P \hat{\phi} J dv = \int_P \dot{\hat{\phi}} J + \hat{\phi} \dot{J} dv$$

$$= \int_P \dot{\hat{\phi}} J dv + \int_P \hat{\phi} J \text{div } \underline{\chi} dv$$

$$\int_P \dot{\hat{\phi}} J dv + \int_{\partial P} \hat{\phi} J \underline{\chi} \cdot \underline{n} dv = \int_P \dot{\hat{\phi}} dv + \int_{\partial P} \hat{\phi} J \underline{\chi} \cdot \underline{n} da$$

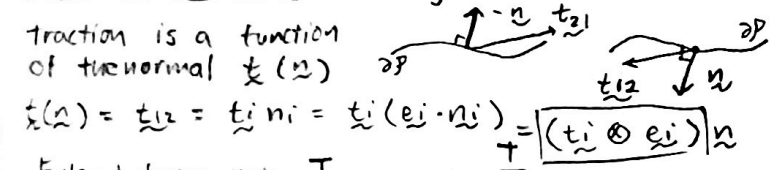
Important side relation: $\underline{j} = J \text{div } \underline{\chi}$

RIGID BODY $\underline{F}(\underline{x}, t) = \underline{Q}(t)$ ★

only a fct of time $t!$ $\Rightarrow \underline{x} = \underline{Q}(t) \underline{x} + \underline{c}(t)$

$$\underline{H} = \underline{J} \underline{\omega} \quad \underline{M} = \underline{\omega} \times \underline{J} \underline{\omega} + \underline{J} \dot{\underline{\omega}}$$

CAUCHY STRESS Cauchy lemma: $t_{12} = -t_{21}$



Euler balance using $\underline{T}$

$$\int_P \rho \dot{\underline{x}} dv = \int_P \rho \underline{b} dv + \int_{\partial P} \underline{t} da = \int_P \rho \underline{b} dv + \int_{\partial P} \underline{T} \underline{n} da$$

$$= \int_P \rho \underline{b} dv + \int_P \text{div } \underline{T} dv \Rightarrow \int_P (\rho \dot{\underline{x}} - \rho \underline{b} - \text{div } \underline{T}) dv = 0$$

localization thm

NORMAL & SHEAR

$$\sigma = \underline{t} \cdot \underline{n} = \underline{T} \underline{n} \cdot \underline{n}$$

$$\tau = \underline{s} \cdot \underline{t} = \underline{s} \cdot \underline{T} \underline{n}$$

$$\underline{T} \underline{s} = (\underline{T} - \underline{n} \otimes \underline{n}) \underline{T} \underline{n}$$

FLUIDS

Non-newtonian: $\underline{T} = g_0 \underline{I} + g_1 \underline{D} + g_2 \underline{D}^2$

where $g_i = g(I_1(\underline{D}), I_2(\underline{D}), I_3(\underline{D}), \rho)$

Newtonian: $\underline{T} = -P(\rho) \underline{I} + \lambda(\rho) (\text{tr } \underline{D}) \underline{I} + 2\mu(\rho) \underline{D}$

Incompressible: $\underline{T} = P(\underline{x}, t) \underline{I} + 2\mu(\rho) \underline{D}$

(div $\underline{\chi} = 0$)

Inviscid: $\underline{T} = -P(\rho) \underline{I}$

EIGENVALS & VECS (AKA principal values / vector)

For a symmetric tensor $\underline{A}$, all 3 eigenvals are real & all 3 eigenvcs are orthogonal to each other, thus spectral decomposition is OK.

$$\underline{A} = \sum_{i=1}^3 \lambda_i \underline{u}_i \otimes \underline{u}_i$$

// note, NOT i notation b/c 3 subscripts

Divergence Thm. $\oint \text{div } \underline{F} dv = \oint \underline{F} \cdot \underline{n} da$

Scalar: $\int_P \phi_{,i} dv = \int_{\partial P} \phi n_i da$

Vector: $\int_P v_{i,j} dv = \int_{\partial P} v_i n_j da$

Tensor: $\int_P A_{ij,j} dv = \int_{\partial P} A_{ij} n_j da$

BALANCE LAWS localization thm

Mass: $\frac{d}{dt} \int_P \rho dv = 0 \Rightarrow \int_P (\dot{\rho} + \rho \text{div } \underline{\chi}) dv = 0$

$$\dot{\rho} + \rho \text{div } \underline{\chi} = 0 \Rightarrow \rho' + \text{div}(\rho \underline{\chi}) = 0$$

ΣM & ΣF :

$$\underline{E} = \underline{E}_b + \underline{E}_c = \int_P \rho \underline{b} dv + \int_P \underline{x} da$$

$$\underline{M}_0 = \underline{x} \times \underline{E}_b + \underline{x} \times \underline{E}_c = \int_P \rho (\underline{x} \times \underline{b}) dv + \int_P (\underline{x} \times \underline{t}) da$$

Euler's balance equations: (glorified $\Sigma F = ma$)

$$\underline{\dot{G}} = \underline{E} \Rightarrow \int_P \rho \dot{\underline{x}} dv = \underline{E}$$

$$\underline{\dot{H}} = \underline{M}_0 \Rightarrow \int_P \rho (\underline{x} \times \dot{\underline{x}}) dv = \underline{M}_0$$

Center of Mass: $\underline{\bar{x}}$

① $\underline{G} = m \underline{\bar{x}}$ ② $\underline{E} = m \underline{\ddot{\bar{x}}}$ ③ $\underline{H}_0 = \underline{\bar{x}} \times \underline{G} + \underline{H}$ ④ $\underline{M}_0 = \underline{\bar{x}} \times \underline{E} + \underline{M}$

SUPERPOSED TRANSFORMS (Keys are $\underline{Q}$ & $\underline{K}$)

Scalars

$$\underline{J}^+ = \underline{J}$$

$$\rho^+ = \rho$$

$$\rho \underline{K}^+ = \rho \underline{K}$$

$$ds^+ = ds$$

$$da^+ = da$$

$$dv^+ = dv$$

Tensors

$$\underline{E}^+ = \underline{Q} \underline{E} \underline{K}^T$$

$$\underline{R}^+ = \underline{Q} \underline{R} \underline{K}^T$$

$$\underline{U}^+ = \underline{K} \underline{U} \underline{K}^T$$

$$\underline{V}^+ = \underline{Q} \underline{V} \underline{Q}^T$$

$$\underline{C}^+ = \underline{K} \underline{C} \underline{K}^T$$

$$\underline{B}^+ = \underline{Q} \underline{B} \underline{Q}^T$$

$$\underline{L}^+ = \underline{Q} \underline{L} \underline{Q}^T + \underline{\Omega}$$

$$\underline{D}^+ = \underline{Q} \underline{D} \underline{Q}^T$$

$$\underline{W}^+ = \underline{Q} \underline{W} \underline{Q}^T + \underline{\Omega}$$

$$\underline{T}^+ = \underline{Q} \underline{T} \underline{Q}^T$$

$$\underline{P}^+ = \underline{Q} \underline{P} \underline{K}^T$$

$$\underline{S}^+ = \underline{K} \underline{S} \underline{K}^T$$

Vectors

$$\underline{t}^+ = \underline{Q} \underline{t}$$

Other identities

$$\underline{F}^+ = \underline{R}^+ \underline{U}^+ = \underline{V}^+ \underline{R}^+ \quad \underline{F}^+ = \underline{L}^+ \underline{F}^+$$

$$\underline{L}^+ = \underline{D}^+ + \underline{W}^+$$

KINEMATIC CONSTRAINTS

general form: $g(\underline{A}) = 0$

where $\underline{A}$ is any tensor & g any fct.

① Incompressible:

$$J = 1 \quad dv = dV \quad \det(\underline{C}) = \det(\underline{F}^T \underline{E})$$

$$g(\underline{C}) = \det(\underline{C}) - 1 \quad J^2 = 1$$

② Inextensibility: $\lambda = 1$

$$\lambda^* = \underline{M} \cdot \underline{C} \underline{M} = 1 \quad g(\underline{C}) = \underline{M} \cdot \underline{C} \underline{M} - 1$$

③ Rigidity $\underline{F} = \underline{Q}(t)$

$$\underline{C} = \underline{F}^T \underline{F} = \underline{Q}^T \underline{Q} = \underline{I} \quad g(\underline{C}_{AB}) = \underline{C}_{AB} - \delta_{AB}$$

TABLE OF VALUES

Value	Name	Type	Intuition	FORMULAS
$\underline{x}$	position vector in referential	Lagrange vector	vector indicating position	$\underline{x} = \underline{x}^{-1}(\underline{\chi}, t)$
$\underline{\chi}$	position vector in current config	Eulerian vector	" " "	$\underline{\chi} = \underline{\chi}(\underline{x}, t) = \underline{\chi}_K(\underline{x}, t)$
$\underline{X}$	point in refer.	Point	Material descriptor	
$\underline{\chi}, \underline{\chi}_K$	NA	Motion	set of equations mapping $\underline{X} \leftrightarrow \underline{\chi} \leftrightarrow \underline{x}$	
$\hat{\underline{v}}, \hat{\underline{a}}$	Lagrangian vel / accl	vectors	Lagrangian uses reference config. at time $t = t_0$, (follow the fluid)	
$\tilde{\underline{v}}, \tilde{\underline{a}}$	Eulerian vel / accl	vectors	Eulerian uses current config at time $t = t$, (follow a fixed pt in space)	
$\mathcal{B}$	Body	object		
$\dot{}$	scalar material derivative	operator	change of a quantity at a certain point from current config = internal changes + what comes in/out	$\dot{f} = \frac{\partial}{\partial t} \bar{f}(\underline{x}, t) = \frac{\partial}{\partial t} \hat{f}(\underline{X}, t) = \frac{\partial}{\partial t} \tilde{f}(\underline{x}, t) + \nabla \tilde{f} \cdot \underline{v}$
$\underline{d}$	vector material derivative	operator	" but for vectors	$\underline{d} = \frac{\partial}{\partial t} \tilde{\underline{u}}(\underline{x}, t) = \frac{\partial}{\partial t} \hat{\underline{u}}(\underline{X}, t) = \frac{\partial}{\partial t} \underline{u}(\underline{x}, t) + (\nabla \underline{u}) \underline{v}$
$\tilde{f}'$	derivative in current config	operator	used for simplicity of previous 2 expressions	$\tilde{f}' = \frac{\partial}{\partial t} \tilde{f}(\underline{x}, t)$
$\underline{L}$	spatial velocity gradient	tensor	used for simplicity again	$\underline{L} = \nabla \underline{v} = \frac{\partial \underline{v}_i}{\partial x_j} \underline{e}_i \otimes \underline{e}_j$
$\underline{F}$	DEFORMATION GRADIENT	tensor	transformation tensor between ref $\rightarrow$ curr. configuration vectors ($\underline{X} \rightarrow \underline{\chi}$)	$\underline{F} = \text{Grad } \underline{\chi}_K$ $d\underline{\chi} = \underline{F} d\underline{X}$ $\underline{F} = \underline{\chi}_{i,A} (\underline{e}_i \otimes \underline{E}_A)$ $\text{Grad } \underline{u} = (\nabla \underline{u}) \underline{F} = \underline{u}_{i,A} \underline{e}_i \otimes \underline{E}_A$
$\underline{J}$	JACOBIAN of DEF. GRAD.	scalar	conversion factor when using $\underline{F}$ to map $\underline{X} \rightarrow \underline{\chi}$	$\underline{J} = \det \underline{F}$ $dV = \underline{J} d\underline{V}$ $\underline{j} = \underline{J} \underline{d}$
$\underline{F}^*$	cotactor of $\underline{F}$	tensor	used in Nanson's formula	$\underline{n} dA = \underline{F}^* \underline{N} dA$ $\underline{F}^* = \underline{J} \underline{F}^{-T}$
$\underline{R}$	orthogonal rotation tensor	tensor	orthogonal Rotator!	$\underline{F} = \underline{R} \underline{U} = \underline{V} \underline{R}$
$\underline{U}$	right stretch tensor	Lagrangian tensor	symmetric, P-D scalar in ref config (pre-rotation)	
$\underline{V}$	left stretch tensor	Eulerian tensor	symmetric, P-D scalar in curr. config (post-rotation)	
$\underline{C}$	Right Cauchy-Green deformation tensor	tensors	symmetric, P-D	$\underline{C} = \underline{F}^T \underline{F} = \underline{U}^2$
$\underline{B}$	Left " " " "	tensors	symmetric, P-D	$\underline{B} = \underline{F} \underline{F}^T = \underline{V}^2$
$\underline{E}$	Lagrangian strain tensor	tensors	motivated by finding arclength differences between $\underline{x}$ & $\underline{\chi}$ ($ds^2 - dS^2$)	$\underline{E} = \frac{1}{2} (\underline{C} - \underline{I})$
$\underline{e}$	Eulerian " " " "	tensors		$\underline{e} = \frac{1}{2} (\underline{B} - \underline{I})$
$\underline{M}$	Local stretch	scalar	ratio of arclengths s & S in ref and curr configs	$\underline{M} \underline{M} = \underline{F} \underline{M}$ $\underline{M} = \underline{X}'(S)$ $\underline{m} = \underline{\chi}'(s)$
$\underline{M}, \underline{m}$	simplifiers	vectors	derivative of arclengths for	$\underline{M}^2 = \underline{M} \cdot \underline{C} \underline{M}$ $\underline{M}^{-2} = \underline{m} \cdot \underline{B} \underline{m}$
$\underline{D}$	Rate of deformation	tensor	See name*	$\underline{D} = \text{Sym}(\underline{L}) = \frac{1}{2} (\underline{L} + \underline{L}^T)$ $\underline{L} = \underline{D} + \underline{W}$
$\underline{W}$	vorticity	tensor	See name*	$\underline{W} = \text{Skw}(\underline{L}) = \frac{1}{2} (\underline{L} - \underline{L}^T)$
$\underline{\omega}$	vorticity vector	vector	vector representation of vorticity	$\underline{W} \underline{u} = \underline{\omega} \times \underline{u}$ $\underline{\omega} = \frac{1}{2} \epsilon_{ijk} W_{kj}$
$\underline{G}$	Linear momentum	vector	integral of all mass in dV multiplied by $\underline{v}$	$\underline{G} = \int \rho \underline{\chi} dV$
$\underline{H}_0$	Angular momentum	vector	" " " " $\underline{x} \times \underline{\chi}$	$\underline{H}_0 = \int \rho (\underline{x} \times \underline{\chi}) dV$
$\underline{F}_b, \underline{b}$	Body Forces	vectors	forces acting on volumes (bodies)	$\underline{F}_b = \int \rho \underline{b} dV$ $\underline{b} \triangleq$ body force / mass
$\underline{F}_c, \underline{t}$	Contact Forces AKA traction	vectors	forces acting on surfaces (traction)	$\underline{F}_c = \int \rho \underline{t} dA$ $\underline{t} \triangleq$ contact force / unit area

Value	Name	Type	Intuition	Formulas
$\bar{x}$	Center of Mass	vector	sum of infinitesimal masses weighted by x	$\bar{x} = \frac{1}{M} \int \rho x dV$
$Q(t)$	position - ind. deformation	tensor	used for rigid bodies where transformation matrix ind. of pos.	$\underline{F}(x, t) = Q(t)$ $Q Q^T = I$
$\underline{\Omega}$	???	tensor	it is skew-symmetric	$\underline{\Omega} = \dot{Q} Q^T$
$\underline{I}$	inertia tensor	tensor	tensor of rotational inertias (typically symmetric)	$\underline{I} = \int \rho [(\underline{x} \cdot \underline{x}) \underline{I} - \underline{x} \otimes \underline{x}] dV$
$\underline{\pi}$	inertia vector	vector	vector of $[I_{xx}, I_{yy}, I_{zz}]^T$	$\dot{\underline{\pi}} = \dot{Q} Q^T \underline{\pi} = \underline{\Omega} \underline{\pi} = \underline{\omega} \times \underline{\pi}$
$\underline{I}$	Cauchy Stress Tensor	tensor	tensor of stresses in σ_{ij} SYMMETRIC $\text{div } \underline{I} + \rho \underline{b} = \rho \underline{\ddot{x}}$ $\underline{I} = \underline{I}^T$	$\underline{I} = t_i \otimes \underline{e}_i$ $T_{ij} = \underline{e}_i \cdot \underline{t}_j \Rightarrow t_j = T_{ij} \underline{e}_i$
$\sigma, \tau, \dots$	traction components	vector/scalar	breaks $\underline{t}$ down into normal & shear	$\underline{t} = \sigma \underline{n} + \tau \underline{s}$
$\underline{P}$	1st Piola-Kirchhoff Stress	tensor	forces on x in curr. per unit A in referential config	$\underline{P} = \underline{I} \underline{F}^* = \underline{J} \underline{I} \underline{F}^{-T}$
$\underline{P}$	1st Piola-Kirchhoff traction	vector	??	$\underline{p} = \underline{P} \underline{N}$ $\text{div } \underline{P} + \rho \underline{b} = \rho \underline{\ddot{x}}$
$\underline{S}$	2nd Piola-Kirchhoff Stress	tensor	LITERALLY NOTHING	$\underline{P} = \underline{F} \underline{S}$
K	Kinetic energy	scalar		$K = \int \frac{1}{2} \rho \underline{\dot{x}} \cdot \underline{\dot{x}} dV$
S	Stress Power	scalar	* See name	$\frac{d}{dt} K + S = P$ $S = \int \underline{I} \cdot \underline{D} dV = \int \underline{I} \cdot \underline{L} dV$
P	Power of Applied Forces	scalar		$P = \int \rho \underline{b} \cdot \underline{x} dV + \int_{\partial V} \underline{t} \cdot \underline{x} dA$
U	internal energy	scalar		$\rho \dot{\underline{e}} = \underline{I} \cdot \underline{L} - \text{div } \underline{q} + \rho r$
S	entropy?	scalar	* See name	$\underline{e}$: internal energy U per unit mass
H	heating rate	scalar		$\underline{h}$: inward scalar heat flux r : heating rate H " " "
$Q(t)$	observer transformer	tensor	transforms one curr config x to another x^*	$x^* = Q x + c$
$\underline{K}$	Observer transformer	tensor	Q (@ some reference time)	$\underline{K} = Q(t_0)$ $x^* = \underline{K} x + c(t_0)$
$\lambda(\rho), \mu(\rho)$	viscosity	scalars	experimentally determined	* check FLUIDS section
$P(\rho)$	Pressure	scalar	pressure as a function of something	* check FLUIDS

GENERAL PROBLEM SOLVING STRATEGY

- Obtain $\underline{F}$ using $\underline{F} d\underline{x} = d\underline{x}$
- Using $\underline{F} \rightarrow \underline{C} \underline{B} \underline{R} \underline{U} \underline{V}$
- Obtain $\underline{L}$ using $\dot{\underline{F}} \underline{F}^{-1}$ or $\underline{L} d\underline{x} = d\underline{\dot{x}}$
- Using $\underline{L} \rightarrow \underline{D} \underline{W} \underline{\omega}$
- Find $\underline{I}$ using $\underline{C}, \underline{B}, \underline{D}$
- Solve Dif eq $\text{div } \underline{I} + \rho \underline{b} = \rho \underline{\ddot{x}}$ if needed. (USEFUL)

Tensor product properties

$$\begin{aligned}
 (\underline{u} \otimes \underline{v}) &= (\underline{x} \otimes \underline{u})^T \\
 (\underline{x} + \underline{w}) \otimes \underline{u} &= \underline{x} \otimes \underline{u} + \underline{w} \otimes \underline{u} \\
 &= (\underline{x} \otimes (\underline{u} + \underline{w}))^T \\
 c(\underline{x} \otimes \underline{u}) &= (c\underline{x}) \otimes \underline{u} = \underline{x} \otimes (c\underline{u}) \\
 \text{M12 Answers } \textcircled{1} \text{ trivial} \\
 \textcircled{2} \underline{L} &= (r\dot{s})' \underline{e}_\theta \otimes \underline{e}_r - f(\underline{e}_r \otimes \underline{e}_\theta) \\
 2\underline{D} &= r\dot{s}'(\underline{e}_r \otimes \underline{e}_\theta + \underline{e}_\theta \otimes \underline{e}_r) \\
 2\underline{W} &= (r\dot{s}' + 2s)(\underline{e}_r \otimes \underline{e}_\theta - \underline{e}_\theta \otimes \underline{e}_r) \\
 \textcircled{3} \text{ for } f &= A + B/r^2 \quad A = (b^2 w_b - a^2 w_a) / (b^2 - a^2) \quad B = a^2 b^2 (w_a - w_b) / (b^2 - a^2)
 \end{aligned}$$

CYLINDRICAL COORDS

$$\begin{aligned}
 \underline{e}_r(\theta) &= \underline{e}_1 \cos \theta + \underline{e}_2 \sin \theta \\
 \underline{e}_\theta(\theta) &= -\sin \theta \underline{e}_1 + \cos \theta \underline{e}_2 \\
 \underline{k} &= \underline{e}_3
 \end{aligned}$$

Gradient in cylindrical

$$\begin{aligned}
 \nabla \phi \cdot d\underline{x} &= d\phi \\
 &= \frac{\partial \phi}{\partial r} dr + \frac{\partial \phi}{\partial \theta} d\theta + \frac{\partial \phi}{\partial z} dz \\
 &= \frac{\partial \phi}{\partial r} (\underline{e}_r \cdot d\underline{x}) + \frac{\partial \phi}{\partial \theta} (\underline{e}_\theta \cdot d\underline{x}) \\
 \nabla \phi &= \frac{\partial \phi}{\partial r} \underline{e}_r + \frac{\partial \phi}{r \partial \theta} \underline{e}_\theta + \frac{\partial \phi}{\partial z} \underline{k}
 \end{aligned}$$

$$\frac{\partial \phi}{\partial \theta} = \underline{e}_\theta \quad \frac{\partial \phi}{\partial r} = \underline{e}_r$$

Differentials $d\underline{x} \rightarrow d\underline{r}, d\underline{\theta}, d\underline{k}$

$$\text{let } \underline{x} = r \underline{e}_r + z \underline{k}$$

$$d\underline{x} = d(r \underline{e}_r + z \underline{k})$$

$$d\underline{x} = dr \underline{e}_r + r d\underline{\theta} + dz \underline{k}$$

follows

$$\underline{e}_r \cdot d\underline{x} = dr$$

$$\underline{e}_\theta \cdot d\underline{x} = r d\theta$$

$$\underline{k} \cdot d\underline{x} = dz$$

APPROACH TO FLUIDS

- Begin with Cauchy Stress $\underline{I} = P(\underline{x}, t) \underline{I} + 2\mu \underline{D}$ for incompress.
- set up balance equation $-\nabla P + 2\mu \text{div } \underline{D} + \rho \underline{b} = \rho(\underline{x}' + (\nabla \underline{x}) \underline{x})$
- find $\underline{L}$ using $\underline{x} = r \underline{e}_r(\theta) + z \underline{k}$ or any position coord system and use $\underline{L} d\underline{x} = d\underline{\dot{x}}$, with $\underline{L}$ find $\underline{D}, \underline{W}$
- start solving for terms ($\text{div } \underline{D}, \underline{L} \underline{x}, -\nabla P$) in component form & coefficient - match the components.